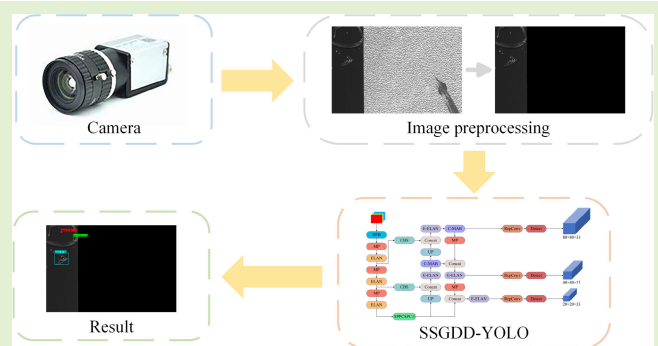


SSGDD-YOLO: Multiscale Feature Fusion and Multiattention-Based YOLO for Smartphone Screen Glass Defect Detection

Ping Wu¹, Haote Zhou, Yicheng Yu, Zengdi Miao, Qianqian Pan¹, Xi Zhang¹, and Jinfeng Gao¹

Abstract—Surface defect detection is essential for ensuring the product quality of smartphone screen glass. In this work, a smartphone screen glass defect detection model based on an enhanced YOLOv7 framework with multiscale feature fusion and multiattention, named SSGDD-you only look once (YOLO) is proposed. In the developed SSGDD-YOLO model, the branch fusion block (BFB) is integrated low-level features from multiple scales through parallel processing, to enhance the details in lower level features for minimizing the information loss as less as possible. Furthermore, the SPPCSPC module of the head is improved as the SPPCSPC-I module, by replacing the standard max pooling with local importance-based pooling (LIP) that reflects the importance of features. The developed SPPCSPC-I module allows the network to automatically learn adaptive importance weights of features during downsampling, enhancing the multiscale feature extraction capability with diverse receptive fields. Finally, a contour-mixed attention block (C-MAB) is inserted into the feature fusion section of the network, which enhances spatial and channel information of features to reduce target information loss, improving the representation capability. Experiments are conducted using a challenging real-world defect image dataset gathered from a smartphone screen glass inspection line in an industrial plant. Results show the proposed SSGDD-YOLO model can achieve the highest mAP of 62.46% among all compared methods.

Index Terms— Attention mechanism, defect detection, multiscale feature fusion, smartphone screen glass, YOLOv7.



I. INTRODUCTION

SMARTPHONES have become an indispensable tool for people [1]. As an essential component of smartphones, the quality of the screen glass directly influences display performance, lifespan, and user satisfaction [2]. In the production of smartphone screen glass, various defects may arise, such as color irregularities and physical damage, due to unavoidable disturbances. Accurate detection of these defects is vital for smartphone screen glass production. Given the substantial demand for screen glass, the development of

automated smartphone screen glass defect detection techniques has become an essential component in the production process, crucial for ensuring the quality of screen glass. However, detecting defects in smartphone screen glass is a challenging task. For instance, different scenarios-such as varying lighting conditions, shadows, and the size of defects on smartphone screens-significantly increase the uncertainty and difficulty of detection.

Artificial visual inspection, traditionally used for quality control in industrial products, is often inefficient and susceptible to human fatigue [3]. With the rapid advancement of imaging technologies, machine vision approaches have become indispensable tools in intelligent manufacturing, providing a highly effective alternative to conventional inspection methods [4]. However, these methods typically rely on manual feature extraction and may struggle in complex scenarios, lacking generality and flexibility. To address these challenges, recent efforts have shifted toward deep learning techniques. Deep learning, compared to traditional methods such as support vector machines (SVMs) and k -nearest neighbors (kNN), can automatically learn features and leverage deep image information through neural networks. Convolutional neural network (CNN), in particular, excel at extracting abstract

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latent features from complex objects through convolutional operations, offering significant advantages in handling image and video data. They are increasingly being explored for defect detection tasks, including applications in smartphone screen glass analysis.

Among CNN-based defect detection methods, the semantic segmentation algorithm [5] excels at performing pixel-level detection on images. To address the issues of inconsistent intra-class segmentation results and difficulties in distinguishing interclass segmentation results in semantic segmentation algorithms, Yeung and Lam [6] proposed an attentive boundary-aware transformer framework called ABFormer, which utilizes split-attention boundary-aware fusion (SABF), boundary-aware spatial attention module (BSAM), and boundary-aware channel attention module (BCAM) to segment different types of defects. Yin et al. [7] proposed a new dual-branch surface defect semantic segmentation network (DDSNet), which integrates semantic and boundary information and introduces a global and local feature fusion (GLF) module based on similarity metrics to guide the network in further refining and highlighting the detailed features of defects. However, while semantic segmentation algorithms can accurately identify defect regions at the pixel level, they require significant human and computational resources. In comparison, object detection algorithms have greater advantages.

Object detection can generally be categorized into two types: two-stage object detection [8] and one-stage object detection [9]. In two-stage object detection algorithms [10], [11], the detection process is divided into two stages. First, candidate boxes of objects are generated based on the input image, and features of the objects are extracted. Subsequently, these candidate boxes are refined to further predict and identify the positions and categories of the objects, characterized by high detection accuracy. Chen et al. [12] proposed a fusion model for mobile phone screen defect detection based on Faster R-CNN [13], improving detection accuracy and speed by optimizing feature extraction networks and initial anchor box generation. Wang et al. [14] introduced a mobile phone surface defect (MPSD) detection model based on Faster R-CNN, expanding the dataset while reducing the requirement for sample numbers. However, these algorithms have complex structures and sacrifice detection speed, making them unsuitable for real-time detection. In contrast, one-stage object detection algorithms [15], [16] do not require generating candidate targets beforehand and can directly predict object positions and categories through the network. One-stage object detection algorithms, notably exemplified by you only look once (YOLO) [17], can achieve faster detection speed. YOLO leverages the advantages of the lightweight and efficient Darknet framework, departing from the complex processes of traditional object detection algorithms by transforming the problem into a regression task [18]. This allows it to predict object positions and categories in a single forward pass, achieving a good balance between speed and accuracy. Wang et al. [19] proposed an improved YOLOv7-based algorithm for detecting surface defects on steel materials. By effectively utilizing feature information to minimize information loss,

they replaced the original loss function with the SIOU loss function, thereby enhancing detection accuracy. Yang et al. [20] introduced a novel model for detecting defects in smartphone screens based on YOLOv5-s and Ghostbottleneck. Their approach addresses the challenge of inconsistent defect scales during smartphone screen detection. Kumar et al. [21] utilized YOLOv7 improved with the Adan optimizer to monitor the vibration signals of electric motors. Lin et al. [22] proposed the MobileNetv3-YOLOv4 (MobYOLv4) architecture for real-time deployment on edge computing devices. This architecture uses MobileNetv3 as the backbone and employs depthwise separable convolutions to reduce complexity, achieving a balance between accuracy and detection speed. Wang et al. [23] developed YOLOv7 by combining flexible and efficient training tools with a proposed architecture and a compound scaling method. However, multiscale features are still not fully exploited, resulting in the degradation of defect detection performance.

To address the issue of multiscale features, techniques such as multiscale feature pyramids, dilated convolutions, and attention mechanisms have been developed [24], [25]. It is noted that multiscale feature pyramids are computationally complex, and dilated convolutions suffer from the gridding effect and lack of correlation, whereas attention mechanisms offer a more straightforward and direct application. The attention mechanism plays a crucial role in enabling models to selectively focus on relevant regions containing valuable information amidst background noise [26]. Thus, attention mechanisms have gained significant attention in recent research. Currently, several types of attention mechanisms have been developed, such as channel attention mechanisms, spatial attention mechanisms, and hybrid approaches [27].

Motivated by the above discussions, this work proposes a novel method for smartphone screen glass defect detection, named SSGDD-YOLO, based on YOLOv7-l framework. In the proposed SSGDD-YOLO, an enhanced YOLOv7-l framework featuring improved feature fusion and attention mechanisms is developed to enhance its detection capabilities, thereby improving the overall performance of the defect detection system. Specifically, the lower level features from multiple scales are integrated through parallel processing, augmenting traditional max pooling with local importance-based pooling (LIP) [28], which emphasizes feature significance. The extraction of spatial and channel information is further refined through the incorporation of a contour-mixed attention block (C-MAB), enriching the model's feature extraction capabilities. Collectively, these innovations significantly enhance the model's capacity to extract multiscale features, effectively addressing the challenge of identifying defects that manifest at different scales on smartphone screen glass. The contributions of this work are summarized as follows.

- 1) In defect detection tasks, capturing and retaining detailed information in images is crucial. To achieve this, the branch fusion block (BFB) is designed to enhance the preservation of detailed defect information by fusing shallow features in parallel. By organically combining information extracted from multiple processing paths

within the network structure, the designed BFB not only enhances the model's sensitivity to subtle differences but also bolsters its overall robustness and generalization capability.

- 2) To accurately capture detailed features at multiple scales, the LIP is introduced into the SPPCSPC-I module. This pooling mechanism retains key features by combining receptive fields of different sizes to remain the important information during the multiscale feature extraction process. Specifically, the SPPCSPC-I module can adaptively select the most suitable receptive field sizes for the current task requirements, thereby effectively enhancing the model's ability to capture defect features across various scales.
- 3) To further enhance the model's sensitivity to edge information and reduce information loss, a new component known as the C-MAB is developed by taking the advantages of both global maximum and local average attention mechanisms. In the developed C-MAB, the global maximum operation is employed to highlight the most prominent parts of the image for identifying key areas that may contain defects. And a local average mechanism is utilized to capture important changes for avoiding the information loss.
- 4) The proposed SSGDD-YOLO model, which integrates the BFB, SPPCSPC-I, and C-MAB into the YOLOv7 framework, is validated through experiments on a real-world smartphone screen glass inspection line and compared to existing models.

The remainder of this article is organized as follows. Section II presents a brief overview of the YOLOv7 mode and convolutional block attention module. Section III elaborates on the proposed SSGDD-YOLO model in detail. Section IV demonstrates its feasibility and effectiveness by conducting experiments on a real-world smartphone screen glass inspection line. Finally, Section V presents the conclusions.

II. BRIEF REVIEW OF YOLOV7 AND ATTENTION MECHANISM

A. YOLOv7 Model

The YOLOv7 model is composed of three primary components: the input, the backbone, and the head, as shown in Fig. 1. Unlike YOLOv5 [29], YOLOv7 integrates the Neck component within its Head section [23]. The workflow begins with resizing the input image to a 640×640 RGB format, which is then processed by the backbone. The Head section subsequently performs feature fusion on the outputs from the backbone, resulting in three distinct feature maps. These maps are further processed through RepConv, and after passing through the detection head and undergoing operations such as nonmaximum suppression, they yield the final predictions.

Compared to earlier versions of YOLO models, YOLOv7 offers several advancements. First, it incorporates trainable bag-of-freebies, significantly enhancing detection accuracy without increasing inference costs. Second, it introduces a method to efficiently replace original modules through module re-referencing and dynamically allocate different output layers. Third, it introduces “extend” and “compound scaling”

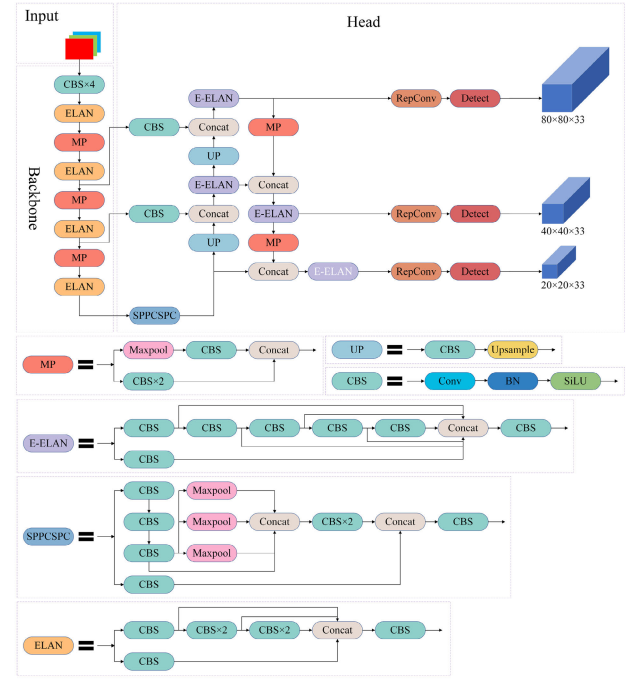


Fig. 1. Architecture of the YOLOv7 model.

techniques, optimizing parameter and computation utilization by 50%, which results in quicker inference and improved detection accuracy. These advantages enable the YOLOv7 model to surpass previous detectors in both speed and accuracy across a spectrum from 5 FPS to 160 FPS. When detecting at speeds exceeding 30 FPS on a GPU V100, YOLOv7's accuracy also exceeds that of other detectors [23].

In performance evaluations across different YOLO variants in image, video, and video surveillance scenarios, although YOLOv8 achieves the fastest detection speed, YOLOv7 realizes the lowest inference time, indicating a trade-off between speed and computational resources despite its higher model complexity. Additionally, YOLOv7 achieves the highest $F1$ score, suggesting a balanced performance between precision and recall [30].

B. Attention Mechanism

The attention mechanism plays a crucial role in enabling models to selectively focus on relevant regions containing valuable information amidst background noise. Initially applied to RNN models for image classification [31], attention mechanisms have gained significant attention in recent research. Today, there are several types of attention mechanisms, each offering unique strengths. Common variants include channel attention mechanisms, spatial attention mechanisms, and hybrid approaches that blend both.

Squeeze and excitation networks (SE-Nets) [32] exemplify the application of attention mechanisms in computer vision, particularly focusing on adjusting feature responses across channels through dynamic recalibration. Selective Kernel networks (SK-Nets) [33] enhance multiscale feature representation by employing multiple parallel convolutional branches with diverse receptive fields, facilitating the selection of

optimal multiscale feature representations. The convolutional block attention module (CBAM) [34], a hybrid approach, combines channel and spatial attention mechanisms. It first applies a channel attention module to compute weighted feature responses, followed by a spatial attention module to refine these results. Coordinate attention (CA) [35] integrates positional information into channel-wise attention calculations, ensuring that the model considers both interchannel dependencies and spatial relationships over long distances. This dual focus enhances detection accuracy and operational efficiency by enabling the model to prioritize important feature channels and spatial patterns within feature maps.

In conclusion, although different attention mechanisms offer distinct advantages, their effectiveness can vary depending on the specific application scenarios. Factors such as data channel correlations significantly influence the outcomes of different attention calculation methods. Therefore, the selection of an appropriate attention mechanism should be tailored to the specific characteristics of the real-world problem at hand.

III. PROPOSED SSGDD-YOLO MODEL

For the purpose of efficiently feature extraction and fusion in smartphone screen glass defect detection, the proposed SSGDD-YOLO model primarily consists of three components: the BFB, the SPPCSPC-I module, and the C-MAB.

A. Branch Fusion Block

Downsampling plays a vital role in capturing higher level features when designing convolutional networks. However, it inevitably sacrifices fine details that are crucial for detecting multiscale defects. Generally, detailed information is most abundant in the shallow layers of the network, closer to the input, where features retain rich, fine-grained data. To address this challenge, the BFB has been developed to effectively merge shallow features, thereby enhancing their representation capability while preserving multiscale detailed information. This strategy ensures the model sensitivity to fine details throughout the network. The architecture of the proposed BFB is illustrated in Fig. 2. As shown in Fig. 2, shallow features are integrated for defect detection.

In the developed BFB, the input structure of YOLOv7 is retained, and it introduces two types of DenseCBS modules. These modules draw inspiration from DenseNet [36], utilizing densely connected shortcuts for feature reuse, as depicted in Fig. 2. Within the DenseCBS module, upon receiving the input, the module first adjusts the number of channels using a 1×1 convolution. Subsequently, a 3×3 convolution compresses the features. The result is then fused with the original input, and this process iterates to achieve feature reuse. Finally, another 1×1 convolution adjusts the channel numbers to produce the output. The DenseCBS module's operation of expanding and compressing features concentrates feature information. Reusing the original features ensures the preservation of their initial distribution, further enhancing feature information. Despite YOLOv7's existing mechanisms for feature reuse, integrating the DenseCBS module compensates for any potential shortcomings in reusing shallow features, thereby augmenting overall feature reuse effectiveness.

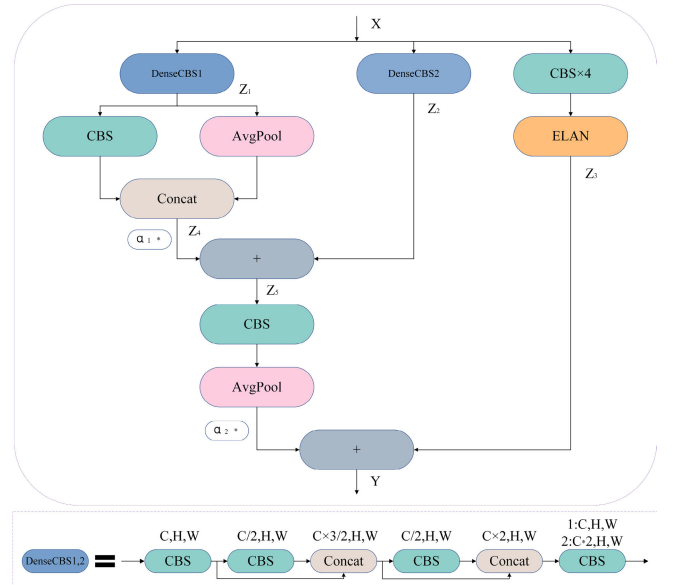


Fig. 2. Architecture of the proposed BFB.

Specifically, assuming the input to the BFB is represented as $\mathbf{X} = (\mathbf{C}, \mathbf{H}, \mathbf{W})$, three feature maps at resolutions of 640×640 , 320×320 , and 160×160 are generated, respectively

$$\begin{aligned} \mathbf{Z}_1 &= \text{DenseCBS1}(\mathbf{X}) \\ \mathbf{Z}_2 &= \text{DenseCBS2}(\mathbf{X}) \\ \mathbf{Z}_3 &= \text{ELAN}(f(\mathbf{X})) \end{aligned} \quad (1)$$

where $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ represent the feature maps at resolutions of 640×640 , 320×320 , and 160×160 , respectively. The function $f(\cdot)$ denotes four consecutive CBS operations. After obtaining feature maps of different resolutions, $\mathbf{Z}_1$ undergoes downsampling through various methods, is fused with $\mathbf{Z}_2$, and introduces learnable parameter $\alpha_1 = (C \times 2, 1, 1)$. This parameter can learn the importance of each channel of the features, enabling the module to selectively retain feature information. The specific process is described by the following equation:

$$\begin{aligned} \mathbf{Z}_4 &= \text{Concat}(\text{CBS}(\mathbf{Z}_1), \text{AvgPool}(\mathbf{Z}_1)) \\ \mathbf{Z}_5 &= \mathbf{Z}_4 * \alpha_1 + \mathbf{Z}_2 \end{aligned} \quad (2)$$

where $\mathbf{Z}_4$ and $\mathbf{Z}_5$ represent the 640×640 feature map after different downsampling operations and the concatenated 320×320 feature map, respectively. The function $\text{Concat}(\cdot)$ denotes fusion in the channel direction, and $\text{AvgPool}(\cdot)$ denotes the average pooling operation. The concatenated feature map $\mathbf{Z}_5$ undergoes a 3×3 convolution for information integration, followed by pooling, and then multiplication with the learnable parameter $\alpha_2 = (C \times 8, 1, 1)$. Finally, it is added to $\mathbf{Z}_3$ to obtain the output $\mathbf{Y}$, as described by the following equation:

$$\mathbf{Y} = \text{AvgPool}(\text{CBS}(\mathbf{Z}_5)) * \alpha_2 + \mathbf{Z}_3. \quad (3)$$

The primary aim of developing this module is to enrich the representation of detailed information in shallow features

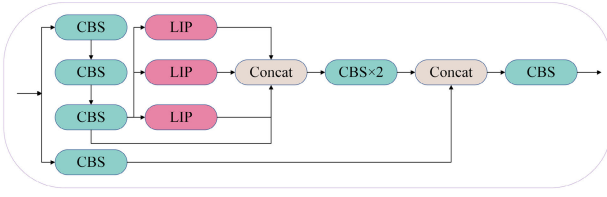


Fig. 3. Architecture of the developed SPPCSPC-I module.

and to maintain comprehensive feature details across different scales, which is particularly crucial for identifying multiscale defects. By employing varied approaches to extract shallow features at different scales, we systematically avoid the loss of feature information. This approach ensures that feature information is preserved in diverse formats through different mechanisms. Additionally, learnable parameters are integrated to regulate the significance of each feature channel. Through the adaptive mechanism, the model can selectively prioritize and retain essential detailed features, thereby enhancing its capacity to accurately identify multiscale defects on smartphone screen glass.

B. SPPCSPC-I Module

In YOLOv7, the SPPCSPC structure integrates two key modules: spatial pyramid pooling (SPP) [37] and cross stage partial connection (CSPC) [38]. Within the SPPCSPC structure, SPP extracts feature representations at multiple scales, followed by CSPC for feature propagation and fusion. This combination effectively manages feature information across multiscales, which is crucial for multiscale defect detection. Thus, to better highlight critical feature details, we employ LIP instead of traditional max pooling, introducing the SPPCSPC-I module, as illustrated in Fig. 3.

LIP [28] is a pooling method that enhances feature extraction by dynamically adjusting importance weights. Like a local attention mechanism, it computes weights via local convolution, autonomously learning the importance of features. Given an input feature $\mathbf{I}$, LIP generates an adaptive importance weight map $F(\mathbf{I})$. This weight map is constructed using exponentiation to ensure nonnegativity and to facilitate optimization

$$F(\mathbf{I}) = \exp(g(\mathbf{I})) \quad (4)$$

where $g(\cdot)$ represents the generated feature importance weights, and $\exp(\cdot)$ denotes the logarithmic module.

During the pooling process, for a feature point (x, y) , we define its neighborhood Ω and normalize the importance weights. Then, the pooled feature is calculated by using the normalized weights for a local sliding window of the original features

$$O_{x',y'} = \frac{\sum_{(\Delta x, \Delta y) \in \Omega} F(\mathbf{I})_{x+\Delta x, y+\Delta y} \mathbf{I}_{x+\Delta x, y+\Delta y}}{\sum_{(\Delta x, \Delta y) \in \Omega} F(\mathbf{I})_{x+\Delta x, y+\Delta y}}. \quad (5)$$

Thanks to its ability to autonomously learn feature importance, similar to local spatial attention, the LIP pooling method enables the SPPCSPC-I module to effectively extract crucial multiscale features during sampling when combined with different receptive field sizes. The SPPCSPC-I module discards

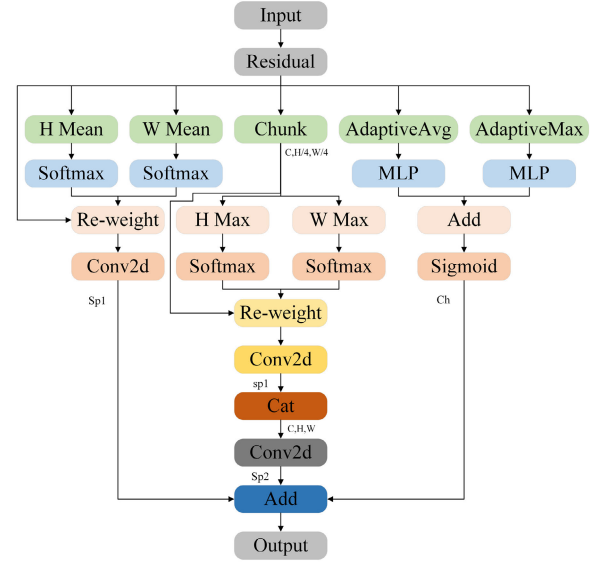


Fig. 4. Architecture of the developed C-MAB.

irrelevant features, enhancing the model's representational and generalization capabilities, and thereby improving multiscale defect detection.

C. Contour-Mixed Attention Block

Since data distribution characteristics can vary across different scenarios, tailoring attention mechanisms to specific scenarios can significantly enhance a model's target detection capabilities. Therefore, in this work, we have developed a customized attention module called the C-MAB for smartphone screen glass defect detection. The C-MAB is illustrated in Fig. 4.

Assuming the input is represented as $\mathbf{X} = (\mathbf{C}, \mathbf{H}, \mathbf{W})$, for spatial attention, we employ global averaging along both the horizontal and vertical dimensions to capture contour details. Additionally, local maximum pooling is adopted to preserve finer features. Global averaging is effective for capturing information related to large-scale defects, as it aggregates feature responses across the entire spatial extent of the input

$$\mathbf{H}' = \text{Softmax}(\text{Mean}_H(\mathbf{X}))$$

$$\mathbf{W}' = \text{Softmax}(\text{Mean}_W(\mathbf{X}))$$

$$\mathbf{Sp1} = \text{Conv2d}(\mathbf{H}' * \mathbf{X} * \mathbf{W}') \quad (6)$$

where $\mathbf{H}'$ is obtained by taking the average of each row of $\mathbf{X}$ and then applying $\text{Softmax}(\cdot)$ function. $\mathbf{W}'$ is derived in a similar manner. Due to the uniform background of the phone's screen glass, defects become prominent because of the high contrast with the background presenting clear contours. Taking the average in this case retains the distribution information of the overall contour, which is then reintegrated into the original input, resulting in $\mathbf{Sp1}$.

Meanwhile, $\mathbf{X}$ is divided into four sub-inputs using $\text{Chunk}(\cdot)$: the top-left $\mathbf{x}_1$, the bottom-left $\mathbf{x}_2$, the top-right $\mathbf{x}_3$, and the bottom-right $\mathbf{x}_4$

$$\mathbf{x}_i = \text{Chunk}(\mathbf{X}), i = 1, 2, 3, 4$$

$$\mathbf{h}'_i = \text{Softmax}(\text{Max}_H(\mathbf{x}_i))$$

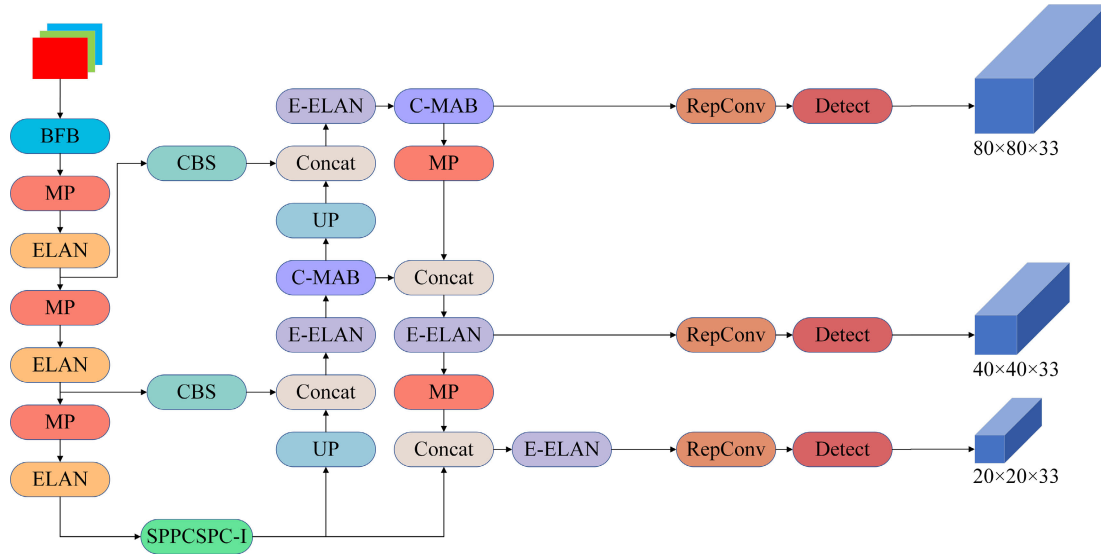


Fig. 5. Scheme of the proposed SSGDD-YOLO model.

$$\begin{aligned} \mathbf{w}'_i &= \text{Softmax}(\text{Max}_W(\mathbf{x}_i)) \\ \mathbf{sp}_i &= \text{Conv2d}(\mathbf{h}'_i * \mathbf{x}_i * \mathbf{w}'_i). \end{aligned} \quad (7)$$

Using the top-left sub-input $\mathbf{x}_1$ as an example, $\mathbf{h}'_1$ is obtained by taking the maximum value of each row of $\mathbf{x}_1$ and then applying $\text{Softmax}(\cdot)$ function. Similarly, $\mathbf{w}'_1$ is derived. Integrating $\mathbf{x}_1$ results in $\mathbf{sp}_1$. The other three sub-inputs undergo the same process to obtain their respective results. In this case, taking the maximum values highlights the prominent local contour information, allowing local pooling to retain details pertinent to small-scale defects.

This dual approach enhances the model's ability to detect defects across different scales efficiently. After extracting the nonlinear transforming horizontal and vertical features, we reintegrate them into the original image to reinforce spatial features and embed positional information. The resulting locally derived spatial attention feature maps are concatenated to preserve resolution and integrate feature details. These maps are then fused with globally derived spatial attention maps to produce the final spatial attention feature maps

$$\mathbf{Sp2} = \text{Conv2d}(\text{Cat}(\mathbf{sp1}, \mathbf{sp2}, \mathbf{sp3}, \mathbf{sp4})). \quad (8)$$

For channel attention, CBAM serves as a baseline. The input undergoes separate global max pooling and global average pooling operations, followed by a two-layer MLP and Sigmoid($\cdot$) function to generate an output

$$\begin{aligned} \mathbf{ch1} &= \text{MLP}(\text{AdaptiveAvg}(\mathbf{X})) \\ \mathbf{ch2} &= \text{MLP}(\text{AdaptiveMax}(\mathbf{X})) \\ \mathbf{Ch} &= \text{Sigmoid}(\mathbf{ch1} + \mathbf{ch2}). \end{aligned} \quad (9)$$

The C-MAB concurrently integrates spatial and channel attention, but treats them independently

$$\mathbf{Y} = \mathbf{Sp1} + \mathbf{Sp2} + \mathbf{Ch}. \quad (10)$$

This unique design combines channel attention feature maps with spatial attention feature maps through addition, further improving the model's ability to extract defect information,

aided by contour extraction techniques. Furthermore, incorporating segmentation and concatenation operations effectively handles defects at various scales, thereby enhancing the overall multiscale feature extraction capabilities.

D. SSGDD-YOLO

By integrating the aforementioned modules and blocks into the YOLOv7-l model framework, the proposed SSGDD-YOLO model achieves superior performance in smartphone screen glass defect detection. The scheme of the proposed SSGDD-YOLO model is illustrated in Fig. 5. As depicted in Fig. 5, the proposed SSGDD-YOLO model integrates the developed BFB, SPPCSPC-I, and C-MAB modules into the YOLOv7-l framework. Initially, the model leverages the BFB within the backbone network to efficiently produce multiscale feature maps, dynamically adjusting the contribution of each resolution's feature map through a channel weight parameter. Subsequently, the SPPCSPC-I module, in conjunction with LIP, is employed to enhance the parallel extraction of multiscale features, selectively preserving essential local and global features to bolster comprehensive and precise detection capabilities. Additionally, the C-MAB is incorporated into the feature enhancement stage to refine information integration, taking into account both spatial and channel dimensions, thereby further augmenting the model's accuracy in detecting and localizing defects across various scales. Specifically, the improvements of the YOLOv7-l model are detailed as follows.

- 1) In the backbone section, the BFB is employed at the input to efficiently generate three types of feature maps with specific resolutions: 640×640 , 320×320 , and 160×160 . Multiscale features are efficiently extracted by introducing dynamic channel weight parameter mechanism. This mechanism finely adjusts the contributions of each resolution's feature map during the fusion of different scale information, notably enhancing the detection capability for small-scale defects. The introduction of the BFB not only strengthens the model's ability to

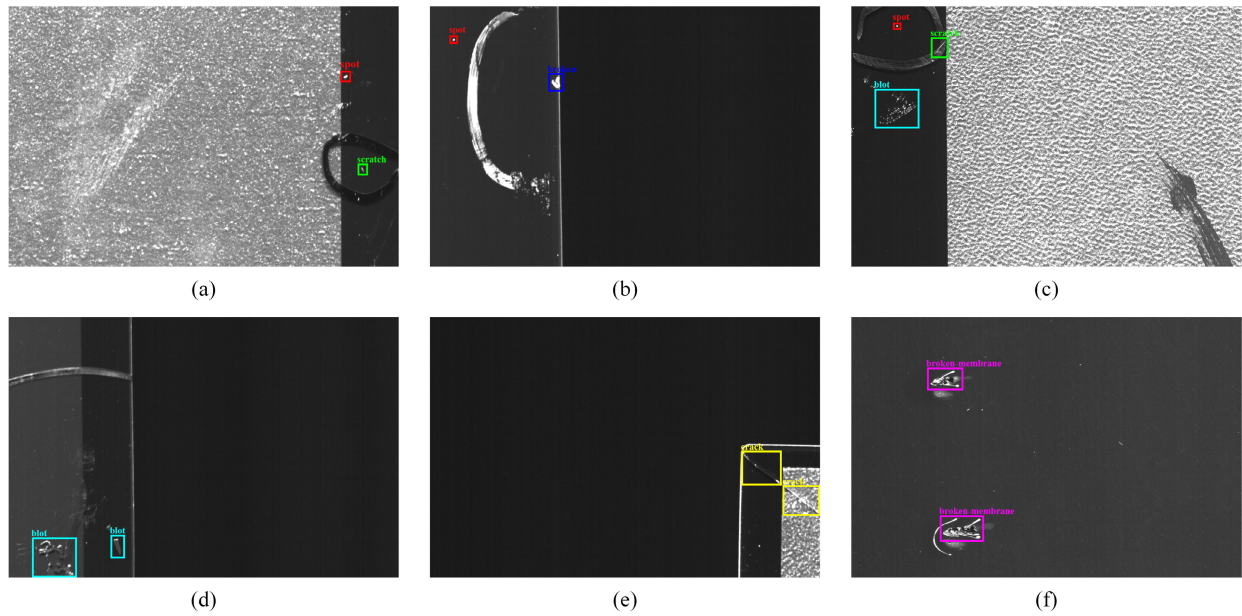


Fig. 6. Illustration of some examples of defects in SSGD. (a) Spot (red) defect and scratch (green) defect. (b) Spot (red) defect and broken (deep blue) defect. (c) Spot (red) defect, scratch (green) defect and blot (light blue) defect. (d) Blot (light blue) defects. (e) Crack (yellow) defects. (f) Broken-membrane (purple) defects.

capture subtle features but also enhances its performance in handling complex scenes with greater ease.

- 2) In the head section, to extend the parallel extraction capability of multiscale features beyond existing multiclass receptive fields, the SPPCSPC-I module is embedded. By introducing LIP, it selectively preserves critical features, retaining rich local features while fully integrating global information. This results in the generation of highly condensed key feature maps, which significantly boost the model's performance in comprehensive and precise detection tasks.
- 3) Moreover, to further optimize the integration of information following feature fusion, the C-MAB is integrated into the feature enhancement stage of the head section, taking into account both spatial positional information and channel information. As a result, defect feature information is extracted from different scales, markedly enhancing the model's performance in accurately identifying and locating targets of various scales in multiscale defect detection.

IV. CASE STUDY

To verify the performance of the proposed SSGDD-YOLO model for defect detection, we employed a smartphone screen glass dataset in this study. The dataset, known as the smartphone screen glass dataset (SSGD), was collected from a real touch screen glass production line [39]. The SSGD contains a variety of defect types across different scales, comprising images captured using line-scan camera which is designed for industry-level information acquisition at fixed workstations. A total of 2504 images with a resolution of 1500×1000 pixels were collected, featuring a black background to minimize ambient light interference.

In this work, we consider six typical defects: spot, broken, scratch, crack, broken-membrane, and blot. Typical defect images are shown in Fig. 6. A total of 956 images is used for training, validation, and testing. The ratio of the training set to the validation set and test set is set as 8 : 1 : 1. Since aluminum foil backgrounds can degrade the detection performance for defects like spots, a Unet network [40] is employed as preprocessing technique. The Unet network can provide flexible labeling for segmenting the aluminum foil background. Some preprocessed images are shown in Fig. 7.

The experiment was conducted on an Ubuntu 18.04 server equipped with a GeForce RTX 3090 GPU and an Intel Xeon Platinum 8255C@2.50GHz CPU. We used the PyTorch framework with Python 3.8 for model training, leveraging GPU acceleration. During testing, both GPU acceleration and CPU were utilized. The input image resolution for the model was set to 640×640 pixels, with a batch size of 8. The optimizer used was stochastic gradient descent (SGD) with a momentum of 0.937, an initial learning rate of 0.01, and a learning rate adjustment strategy employing CosineAnnealingLR. Weight decay was set to 0.0005. Data augmentation was performed using Mosaic and MixUp; there is a 50% probability of using Mosaic data augmentation at each step, and a 50% probability of applying MixUp after Mosaic. The model was trained for a total of 100 epochs, with Mosaic enabled for the first 70% of the epochs. The training loss and validation loss of the model are shown in the Fig. 8. As illustrated in Fig. 8, the SSGDD-YOLO model exhibits robust training performance. The training loss stabilizes after 100 epochs, and the validation loss follows a similar trajectory to the training loss. This pattern suggests that the model has successfully converged, without any apparent signs of overfitting or underfitting. For comparison, several state-of-the-art defect detection methods

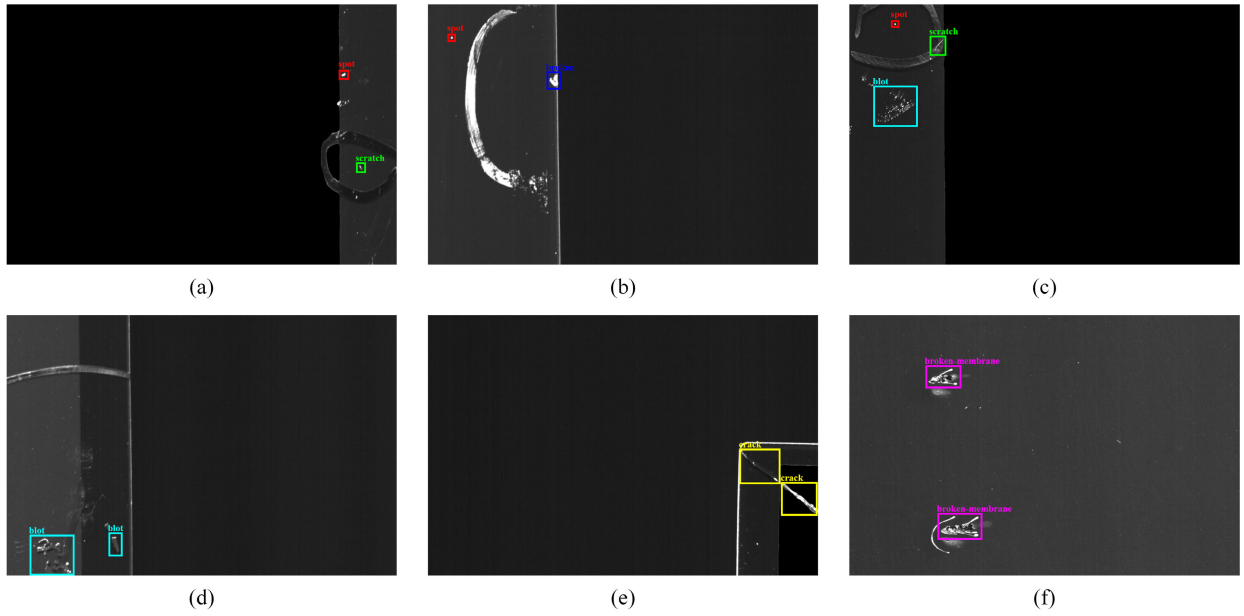


Fig. 7. Illustration of some examples of preprocessed images. (a) Spot (red) defect and scratch (green) defect. (b) Spot (red) defect and broken (deep blue) defect. (c) Spot (red) defect, scratch (green) defect, and blot (light blue) defect. (d) Blot (light blue) defects. (e) Crack (yellow) defects. (f) Broken-membrane (purple) defects.

were employed, including Faster R-CNN [13], RetinaNet [41], SSD [42], YOLOv3 [43], YOLOv4 [44], YOLOv5-l [45], YOLOv8-l [46], YOLOX-l [47], PPYOLOE + -l [48], and RTMDet-l [49]. To evaluate the defect detection performance, we use average precision (AP), mean AP (mAP), the number of parameters (Params), and frames per second (FPS) as evaluation metrics. The calculations for AP and mAP are as follows:

$$\begin{aligned}
 P &= \frac{TP}{TP + FP} \\
 R &= \frac{TP}{TP + FN} \\
 AP &= \int_0^1 P(R) dR \\
 mAP &= \frac{\sum_{i=1}^C AP_i}{C}
 \end{aligned} \quad (11)$$

where P represents precision, R represents recall, TP denotes the number of samples correctly classified as positive, FP denotes the number of samples incorrectly classified as positive, and FN denotes the number of samples incorrectly classified as negative. C represents the number of target categories.

The COCO evaluation metrics, as detailed in [50], are employed to quantitatively assess the detection performance of the attention module. These metrics encompass (AP), which represents the AP across IoU thresholds ranging from 0.50 to 0.95. Additionally, they include AP at specific IoU thresholds: AP_{50} at 0.50, AP_{75} at 0.75. The metrics also distinguish between object sizes: AP_S for small objects (pixel area $< 32 \times 32$), AP_M for medium-sized objects ($32 \times 32 < \text{pixel area} < 96 \times 96$), and AP_L for large objects (pixel area $> 96 \times 96$).

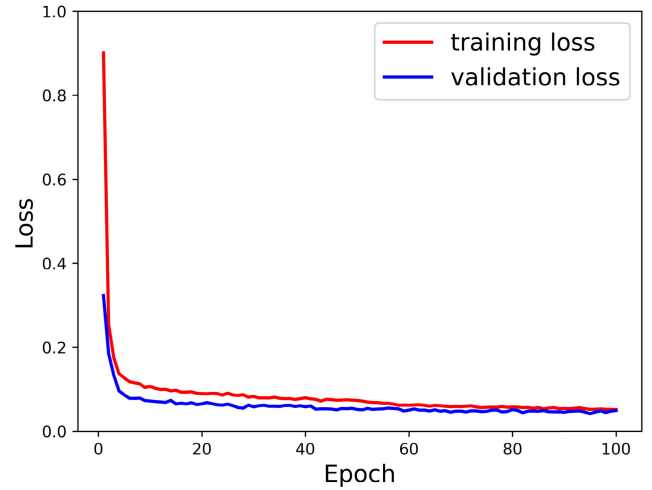


Fig. 8. Training loss and validation loss of the SSGDD-YOLO model.

A. Comparative Results With Different SOTA Methods

In this study, the following state-of-the-art object detection models are employed for comparison: Faster R-CNN [13], RetinaNet [41], SSD [42], YOLOv3 [43], YOLOv4 [44], YOLOv5-l [45], YOLOv8-l [46], YOLOX-l [47], PPYOLOE + -l [48], and RTMDet-l [49].

The proposed SSGDD-YOLO model was evaluated on the SSGD dataset through comparative experiments with other models in the YOLO series, as summarized in Table I. As shown, the YOLO series algorithms exhibit strong proficiency in detecting spot defects, with most achieving APs above 60%, except for YOLOv4 and PPYOLOE + -l. Notably, YOLOv5-l, YOLOv8-l, YOLOX-l, and SSGDD-YOLO perform well in detecting spot, broken, and scratch defects, with

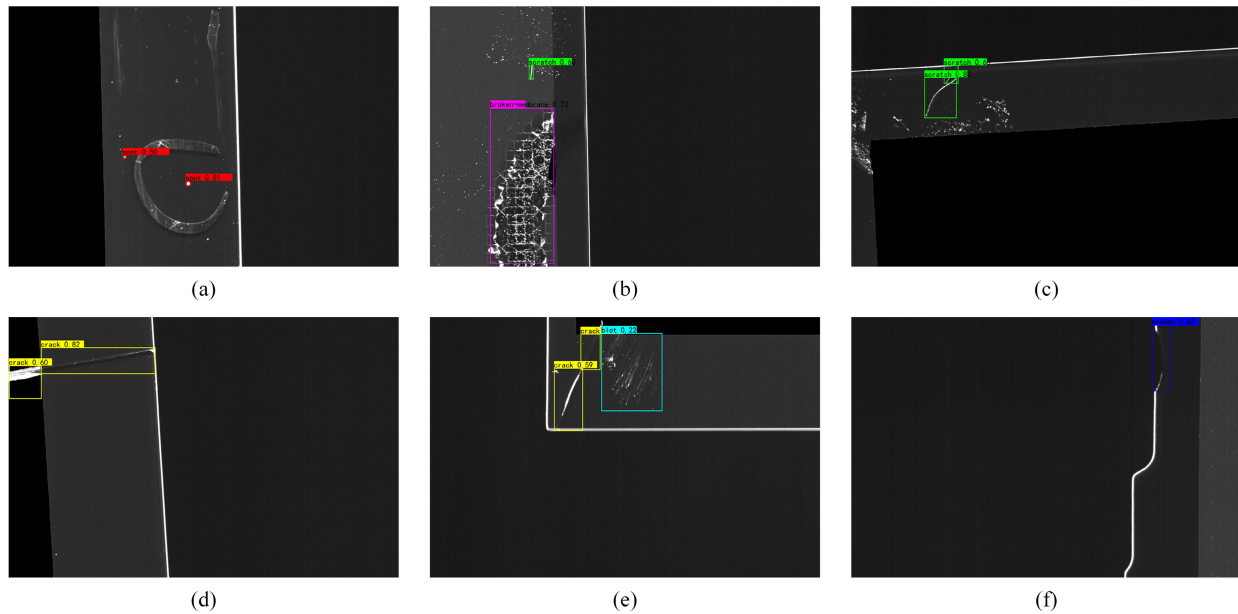


Fig. 9. Example of detection results from the SSGDD-YOLO model. (a) Spot (red) defects. (b) Scratch (green) defect and broken-membrane (purple) defect. (c) Scratch (green) defects. (d) Crack (yellow) defects. (e) Crack (yellow) defects and blot (light blue) defect. (f) Broken (deep blue) defect.

TABLE I
COMPARISON RESULTS OF DIFFERENT MODELS

Model	AP/%						$mAP/\%$	Params/M	FPS
	spot	broken	scratch	crack	broken-membrane	blot			
Faster R-CNN [13]	0.07	46.14	28.13	44.34	28.53	1.61	24.80	136.79	7.61
RetinaNet [41]	0.00	46.12	37.10	49.13	43.33	50.00	37.61	36.43	17.22
SSD [42]	0.81	43.20	21.76	39.18	32.14	4.55	23.61	24.29	48.43
YOLOv3 [43]	61.13	49.15	57.44	46.63	27.78	0.00	40.35	61.55	37.17
YOLOv4 [44]	37.03	0.00	32.27	15.95	17.82	0.00	17.18	63.97	28.81
YOLOv5-l [45]	69.23	68.59	79.75	38.41	33.33	0.00	48.22	46.66	19.83
YOLOv8-l [46]	62.68	66.38	74.45	33.09	33.33	50.00	53.32	43.63	15.82
YOLOX-l [47]	67.39	79.29	80.25	49.26	33.33	6.25	52.63	54.15	16.21
PPYOLOE+ -l [48]	39.27	35.01	54.19	55.43	31.48	34.07	25.42	51.30	31.45
RTMDet-l [49]	38.38	28.04	40.62	55.46	33.00	27.82	45.36	52.26	36.90
SSGDD-YOLO	77.92	74.56	80.76	49.86	41.67	50.00	62.46	48.93	14.81

APs exceeding 62%. SSGDD-YOLO achieves particularly high APs of 77.92% for spot defects and 80.76% for scratch defects, outperforming other models.

However, detecting crack, broken-membrane, and blot defects remains challenging, with APs below 50% for crack and broken-membrane, and below 7% for blot across most models including Faster R-CNN, SSD, YOLOv3, YOLOv4, YOLOv5-l, and YOLOX-l. Nevertheless, SSGDD-YOLO exhibits superior performance with APs of 49.86%, 41.67%, and 50.00% for crack, broken-membrane, and blot, respectively. Overall, although SSGDD-YOLO has a slight increase in Params and a slightly lower FPS compared to other models, trade-off results in achieving the highest mAP of 62.46% among all compared methods.

Example detection results of the SSGDD-YOLO model are presented in Fig. 9. In this Fig. 9, scratch defects, indicated in green, exhibit different sizes in Fig. 9(b) and (c), as do crack defects, marked in yellow, in Fig. 9(d) and (e). Spot defects,

shown in red, generally appear small, as illustrated in Fig. 9(a). On the other hand, broken-membrane defects in purple, blot defects in light blue, and broken defects in deep blue, as seen in Fig. 9(b), (e), and (f), display a variety of sizes. These sizes are not limited to those shown in Fig. 7. As observed in Fig. 9, all defect types are successfully detected.

B. Comparative Results With Different Attention Modules

To further assess its effectiveness, several mainstream attention modules, such as SE, SK, CBAM, CA, PSA [51], and EMA [52], are taken into consideration under the YOLOv7-l (hereafter referred to as YOLOv7) model framework for comparison. The results, summarized in Table II, demonstrate the performance gains achieved by our C-MAB approach in defect detection tasks on the SSGD dataset. According to COCO evaluation metrics, the YOLOv7 + C-MAB model exhibits superior overall performance compared to other combined models. Its AP and AP₅₀ achieve 32.5% and 58.5%,

TABLE II
COMPARISON RESULTS OF DIFFERENT ATTENTION MODULES

Model	$AP/\%$	$AP_{50}/\%$	$AP_{75}/\%$	$AP_S/\%$	$AP_M/\%$	$AP_L/\%$
YOLOv7+SE [32]	28.10	54.60	24.80	42.90	18.50	37.50
YOLOv7+SK [33]	25.10	51.40	20.70	27.00	19.10	32.20
YOLOv7+CBAM [34]	27.40	51.40	26.80	28.70	19.40	33.70
YOLOv7+CA [35]	30.60	57.90	31.00	27.60	21.10	44.60
YOLOv7+PSA [51]	26.90	52.20	25.80	35.80	18.20	34.40
YOLOv7+EMA [52]	27.10	51.10	25.60	47.30	18.30	35.00
YOLOv7+C-MAB	32.50	58.50	30.00	39.00	18.50	47.60

TABLE III
RESULTS OF ABLATION EXPERIMENTS

Model	$AP/\%$						$mAP/\%$
	spot	broken	scratch	crack	broken-membrane	blot	
YOLOv7	72.82	68.13	80.61	52.10	27.78	0.00	50.25
YOLOv7+BFB (only α_2)	79.67	87.44	83.95	51.18	33.33	0.00	55.93
YOLOv7+BFB	74.87	65.86	80.23	42.79	24.46	50.00	56.37
YOLOv7+SPPCSPC-I	76.67	72.86	77.19	42.70	38.89	50.00	59.72
YOLOv7+C-MAB	74.62	76.51	82.50	37.17	36.38	50.00	59.53
YOLOv7+BFB+C-MAB	74.97	71.99	78.10	38.86	48.06	50.00	60.33
SSGDD-YOLO	77.92	74.56	80.76	49.86	41.67	50.00	62.46

respectively, with the AP_{75} metric trailing the best model by only 1.0%.

In terms of the AP_S , AP_M , and AP_L metrics, the YOLOv7 + C-MAB model excels in AP_L , ranks third in AP_S after YOLOv7 + EMA and YOLOv7 + SE, and trails the top-performing YOLOv7 + CA model by 2.6% in AP_M . Overall, while the YOLOv7 + C-MAB model achieves the highest performance only in AP_L , its competitive performance across other metrics indicates its suitability for the SSGD dataset. The significant 10.1% improvement in the AP_L metric highlights the YOLOv7 + C-MAB model's balanced detection capabilities. This suggests that the YOLOv7 + C-MAB model effectively extracts multiscale features and provides comprehensive coverage across various defect sizes, making it well-suited for smartphone screen glass defect detection.

C. Ablation Experimental Results

To study the impact of the BFB, SPPCSPC-I module, and C-MAB on the performance of the SSGDD-YOLO model, an ablation experiment was conducted. The results are presented in Table III, and the data visualization heatmap is shown in Fig. 10. To verify the impact of learnable parameters α_1 and α_2 on the model, additional ablation experiments were conducted by incrementally adding these parameters. According to the experimental results, the detection capability for all defects except crack and blot improves when only the learnable parameter α_2 is effective, achieving a 5.68% increase in mAP. When both learnable parameters α_1 and α_2 are effective, i.e., the complete BFB, although the detection capability for various defects decreases, the detection capability for blot defects significantly improves, resulting in a 0.44% increase

in mAP. Therefore, the learnable parameters α_1 and α_2 are of significant importance to the BFB and even the SSGDD-YOLO model.

Additionally, the YOLOv7 + BFB model demonstrates a notable 6.12% improvement in mAP compared to the baseline YOLOv7 model. The BFB enhances the extraction of multiscale defect details in shallower networks, significantly improving detection capabilities for spot and blot defects, although there is a 9.31% decrease in accuracy for crack defects. The YOLOv7 + SPPCSPC-I model exhibits a substantial 9.47% increase in mAP. The parallel LIPs with varying sizes of convolutional kernels within the SPPCSPC-I module enrich the perception field, enhancing multiscale feature extraction. This improvement translates to enhanced detection performance for spot, broken, broken-membrane, and blot defects, accompanied by a 9.40% decrease in accuracy for crack defects. Similarly, the YOLOv7 + C-MAB model demonstrates a 9.28% mAP improvement. The C-MAB's approach of globally averaging horizontal and vertical features, combined with local maximum pooling, enriches feature representation across different defect scales. Except for crack defects, detection accuracy improves for all other defect types. Moreover, combining the BFB with C-MAB in the YOLOv7 + BFB + C-MAB model enhances detection performance for spot, broken, and broken-membrane defects compared to using BFB alone. Finally, integrating all three modules in the SSGDD-YOLO model yields comprehensive performance improvements. Although detection accuracy for crack defects shows a slight 2.24% decrease, the overall mAP increases significantly by 12.21%. Fig. 10 depicts the data visualization heatmap for various models. By examining

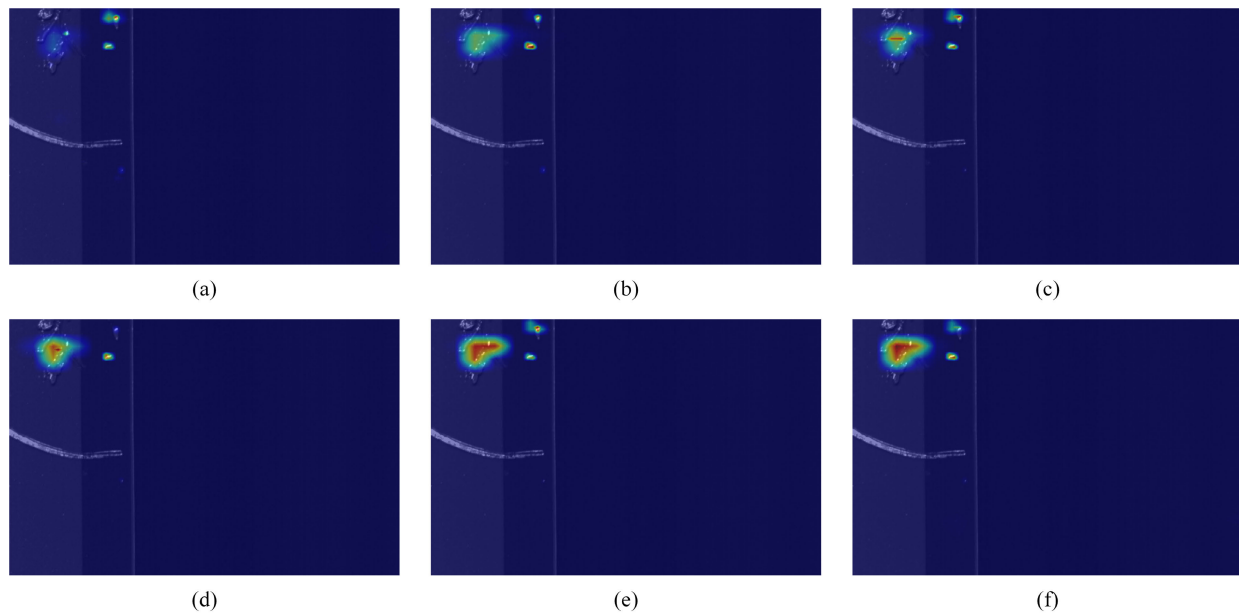


Fig. 10. Illustration of the data visualization heatmap. Higher values are indicated by warmer colors, and lower values are indicated by cooler colors. (a) YOLOv7. (b) YOLOv7 + BFB. (c) YOLOv7 + SPPCSPC-I. (d) YOLOv7 + C-MAB. (e) YOLOv7 + BFB + C-MAB. (f) SSGDD-YOLO.

the heatmap in Fig. 10, one can discern the enhancements contributed by different modules, namely BFB, SPPCSPC-I, and C-MAB, to the model's overall performance. When these modules operate individually, a comparison between the heatmaps of the initial model in Fig. 10(a) and those of the models incorporating each module, Fig. 10(b)–(d), reveals a noticeable darkening of the color in the defect area located at the upper left corner. This darkening signifies higher values, indicating that the model has increased its focus on this region, thereby enhancing its detection capabilities. Furthermore, when two or more modules are combined, the model's detection performance is observed to improve with an increasing number of active modules, as illustrated in Fig. 10(e) and (f). It is evident that by integrating all the developed modules, the SSGDD-YOLO model achieves a more balanced and superior performance across a range of detection metrics.

V. CONCLUSION

In this work, we propose a novel model based on an enhanced YOLOv7 framework to improve feature fusion and attention mechanisms for smartphone screen glass defect detection. Specifically, we have developed the BFB, the SPPCSPC-I module, and the C-MAB. The BFB enhances the model's ability to capture defect features across different scales by merging shallow features. The SPPCSPC module has been extended to the SPPCSPC-I module by introducing LIP to retain critical features and utilizing parallel LIPs with varying convolutional kernel sizes to gather comprehensive contextual information, thereby improving the model's multiscale feature extraction capability. Furthermore, for smartphone screen glass defect detection, the C-MAB has been developed by integrating spatial and channel attention mechanisms to extract hybrid attention features. The SSGDD-YOLO model incorporates these blocks and modules into the YOLOv7 framework.

Experiments conducted on a real smartphone screen glass dataset, collected from a touch screen glass production line, demonstrate the superiority of the proposed SSGDD-YOLO model over other state-of-the-art models in defect detection.

While the SSGDD-YOLO model has shown superior performance compared to other models, there are several avenues for further improvement in future studies. First, the acquisition of defect images is a challenge, suggesting that data augmentation strategies could be effectively utilized to create additional synthetic defect images, which may enhance the model's predictive accuracy. Second, there is a need for a more profound investigation into the integration of high-level and low-level semantic information to sharpen the model's feature extraction capabilities. Third, exploring novel deep learning architectures, such as vision transformer, for defect detection could be a promising direction for future research.

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